

Machine Learning: Quiz 1

(1)

MULTIPLE CHOICE marked out of 1 penalty 0.10 One answer only Shuffle

Which of the following is not an unsupervised learning task?

- a. Clustering
- b. Visualization
- c. Dimensionality reduction
- d. Logistic Regression ✓

(2)

MULTIPLE CHOICE marked out of 1 penalty 0.10 One answer only Shuffle

What type of algorithm would you use to allow a robot to walk in various unknown terrains?

- a. Supervised Learning
- b. Unsupervised Learning
- c. Reinforcement Learning ✓
- d. Semi-Supervised Learning

(3)

MULTIPLE CHOICE marked out of 1 penalty 0.10 One answer only Shuffle

What type of algorithm would you use to segment your customers into multiple groups if you don't know how to define the groups?

- a. Supervised Learning
- b. Unsupervised Learning ✓
- c. Reinforcement Learning
- d. Semi-Supervised Learning

(4)

MULTIPLE CHOICE marked out of 1 penalty 0.10 One answer only Shuffle

Would you frame the problem of spam detection?

- a. Supervised Learning ✓
- b. Unsupervised Learning
- c. Reinforcement Learning

d. Semi-Supervised Learning

(5)

MULTIPLE CHOICE marked out of 1 penalty 0.10 One answer only Shuffle

What is an online learning system?

- a. A system that processes all data at once and updates the model periodically.
- b. A system that can only learn from a fixed dataset.
- c. A system that learns incrementally, adapting to new data as it arrives. ✓
- d. A system that only learns in offline mode.

(6)

MULTIPLE CHOICE marked out of 1 penalty 0.10 One answer only Shuffle

What is out-of-core learning?

- a. A learning algorithm that requires all data to fit in a computer's main memory.
- b. A learning algorithm that processes data in real-time without storing it.
- c. A learning algorithm that handles large datasets by processing them in mini-batches, which do not fit in memory at once. ✓
- d. A learning algorithm that is only suitable for small datasets.

(7)

MULTIPLE CHOICE marked out of 1 penalty 0.10 One answer only Shuffle

What type of algorithm relies on a similarity measure to make predictions?

- a. Supervised Learning
- b. Instance-Based Learning ✓
- c. Model based Learning
- d. Unsupervised Learning

(8)

MULTIPLE CHOICE marked out of 1 penalty 0.10 One answer only Shuffle

Which algorithm is the fastest when dealing with large numbers of instances (large m)?

- a. Normal Equation ✓
- b. Batch Gradient Descent
- c. Stochastic Gradient Descent
- d. Mini-batch Gradient Descent

(9)

MULTIPLE CHOICE marked out of 1 penalty 0.10 One answer only Shuffle

Which algorithm(s) support out-of-core learning?

- a. Normal Equation
- b. SVD
- c. Batch Gradient Descent
- d. Stochastic Gradient Descent ✓

(10)

MULTIPLE CHOICE marked out of 1 penalty 0.10 One answer only Shuffle

Which algorithm requires data scaling before being applied?

- a. Normal Equation
- b. SVD
- c. Batch Gradient Descent ✓
- d. All of the above

(11)

MULTIPLE CHOICE marked out of 1 penalty 0.10 One answer only Shuffle

How many hyperparameters are involved in Batch Gradient Descent?

- a. 0
- b. 1
- c. 2 ✓
- d. More than 2

(12)

MULTIPLE CHOICE marked out of 1 penalty 0.10 One answer only Shuffle

What do model-based algorithms search for?

- a. Hyperparameters
- b. Proximity of points
- c. Target value
- d. Model parameters ✓

(13) **Training Algorithms**

MULTIPLE CHOICE

marked out of 1

penalty 0.10

One answer only

Shuffle

Which linear regression training algorithm can you use if you have a training set with millions of features and if it doesn't fit into memory?

- a. Stochastic Gradient Descent ✓
- b. Batch Gradient Descent
- c. Normal Equation
- d. Singular Value Decomposition (SVD) Approach

(14) **Correct Form of the Normal Equation**

MULTIPLE CHOICE

marked out of 1

penalty 0.10

One answer only

Shuffle

Which of the following is the correct form of the Normal Equation for linear regression?

- a. $\theta = (X^T X)^{-1} X^T y$ ✓
- b. $\theta = X^T (X X^T)^{-1} y$
- c. $\theta = (X X^T)^{-1} X^T y$
- d. $\theta = X^T (X^T X)^{-1} y$

(15) **Correct cost Function in Linear Regression**

MULTIPLE CHOICE

marked out of 1

penalty 0.10

One answer only

Shuffle

Which of the following equations correctly represents the cost function $J(\theta)$ used in linear regression?

- a. $J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$ ✓
- b. $J(\theta) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$
- c. $J(\theta) = \frac{1}{2} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$
- d. $J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$

(16) **Correct Gradient Descent Update Rule**

MULTIPLE CHOICE

marked out of 1

penalty 0.10

One answer only

Shuffle

Which of the following equations correctly represents the Gradient Descent update rule?

- a. $\theta := \theta - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x^{(i)}$ ✓
- b. $\theta := \theta - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$
- c. $\theta := \theta + \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x^{(i)}$
- d. $\theta := \theta - \alpha \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$

(17) **Feature Scaling Impact**

MULTIPLE CHOICE

marked out of 1

penalty 0.10

One answer only

Shuffle

Suppose the features in your training set have very different scales. Which algorithms might suffer from this?

- a. Gradient Descent algorithms ✓
- b. Normal Equation
- c. SVD approach
- d. All of these

(18) **Gradient Descent and Local Minima**

TRUE/FALSE

marked out of 1

Gradient descent can get stuck in a local minimum when training a linear regression model?

- **True** $\rightarrow$ *True*
- **False** ✓ $\rightarrow$ *False*

(19) **Batch Gradient Descent and Convex Problems**

TRUE/FALSE

marked out of 1

If the optimization problem is convex, all Batch Gradient Descent algorithms lead to the same model irrespective of where you start from, provided you let them run long enough?

- **True** ✓ $\rightarrow$ *True*
- **False** $\rightarrow$ *False*

(20)

MULTIPLE CHOICE

marked out of 1

penalty 0.10

One answer only

Shuffle

Suppose you use batch gradient descent and you plot the training error validation error at every epoch. If you notice that the validation error consistently goes up but the training error is not going up, what can you say about the model or fitting?

- a. Overfitting ✓
- b. Low variance
- c. High bias
- d. Underfitting

(21) **Code Output**

MULTIPLE CHOICE

marked out of 1

penalty 0.10

One answer only

Shuffle

Predict the output of the following code snippet:

```
import numpy as np
x = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
print(x[1, 2])
```

- a. 6 ✓
- b. 2
- c. 5
- d. 8

(22) **Code Output**

MULTIPLE CHOICE

marked out of 1

penalty 0.10

One answer only

Shuffle

Predict the output of the following code snippet:

```
import numpy as np
x = np.array([1, 2, 3, 4, 5])
print(x[::2])
```

- a. [1 3 5] ✓
- b. [1 2 3 4 5]
- c. [5 4 3 2 1]
- d. [1 2 3]

(23) Code Output

MULTIPLE CHOICE

marked out of 1

penalty 0.10

One answer only

Shuffle

Predict the output of the following code snippet:

```
import numpy as np
x = np.arange(9).reshape(3, 3)
print(x[1, :])
```

- a. [3 4 5] ✓
- b. [0 1 2]
- c. [6 7 8]
- d. [9 10 11]

(24) Code Output

MULTIPLE CHOICE

marked out of 1

penalty 0.10

One answer only

Shuffle

Predict the output of the following code snippet:

```
import numpy as np
x = np.arange(1,10).reshape(3,3)
print(np.sum(x, axis=1))
```

- a. Sum of elements in each row ✓
- b. Sum of elements in each column
- c. Total sum of elements
- d. Average of elements

(25) Code Output

MULTIPLE CHOICE

marked out of 1

penalty 0.10

One answer only

Shuffle

Predict the output of the following code snippet:

```
import numpy as np
x = np.linspace(0, 2, 5)
print(x)
```

- a. [0. 2. 4.]
- b. [0. 5]
- c. [0. 2.5]
- d. [0. 0.5 1.0 1.5 2.0] ✓

(26) **Code Output**

MULTIPLE CHOICE

marked out of 1

penalty 0.10

One answer only

Shuffle

Predict the output of the following code snippet:

```
import numpy as np
h = np.arange(5).reshape(1, 1, 5)
print(h + [10, 20, 30, 40, 50])
```

- a. `[[[10 21 32 43 54]]]` ✓
- b. `[10 21 32 43 54]`
- c. `ValueError`
- d. `[11 22 33 45 55]`

(27) **Code Output**

MULTIPLE CHOICE

marked out of 1

penalty 0.10

One answer only

Shuffle

Predict the output of the following code snippet:

```
import numpy as np
b = np.arange(24).reshape(4, 6)
print(b[1:2, :])
```

- a. `[[ 6 7 8 9 10 11]]` ✓
- b. `[ 6 7 8 9 10 11]`
- c. `[[ 6 7 8 9 10 11] [12 13 14 15 16 17]]`
- d. `[4 5 6 7]`

Total of marks: 27